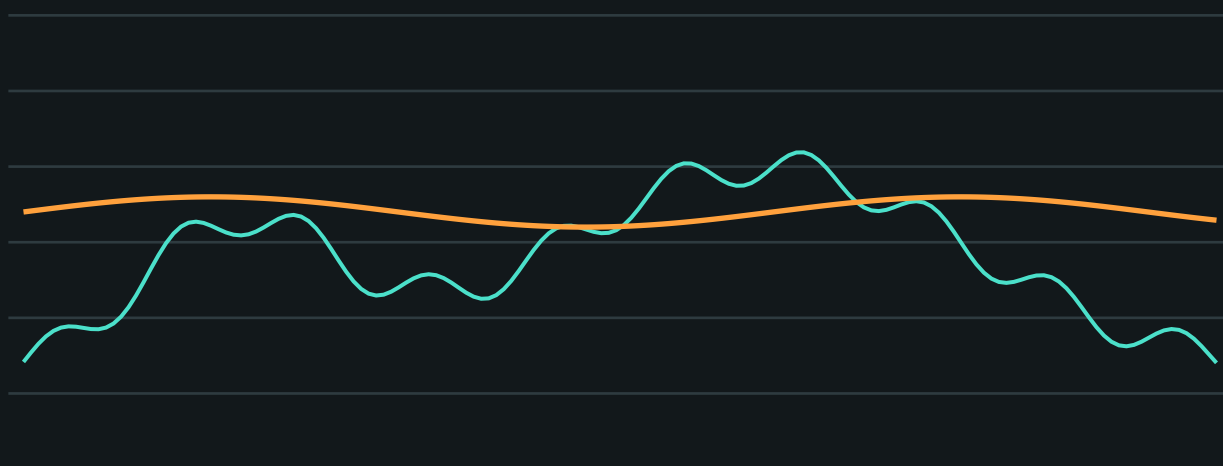


ENGLISH EDITION

TB Spectral Transient Shaper

User manual for the spectral attack/sustain shaper plugin

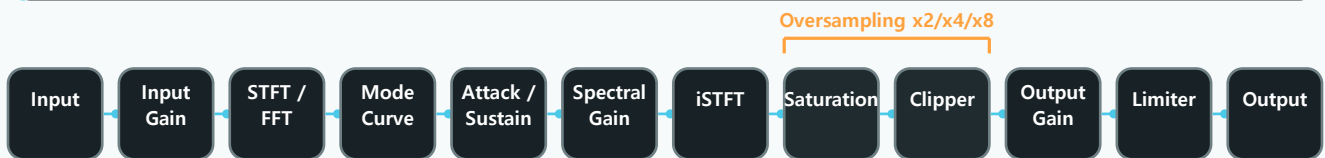


Implementation baseline: 2026-06-30

Overview

TB Spectral Transient Shaper analyzes the incoming signal with an STFT/FFT path, separates frequency bins into Attack and Sustain activity, then applies independent gain to the detected regions. The global Saturation, Clipper, and Limiter modules sit after the transient shaping stage and shape the final tone and level.

Processing Order



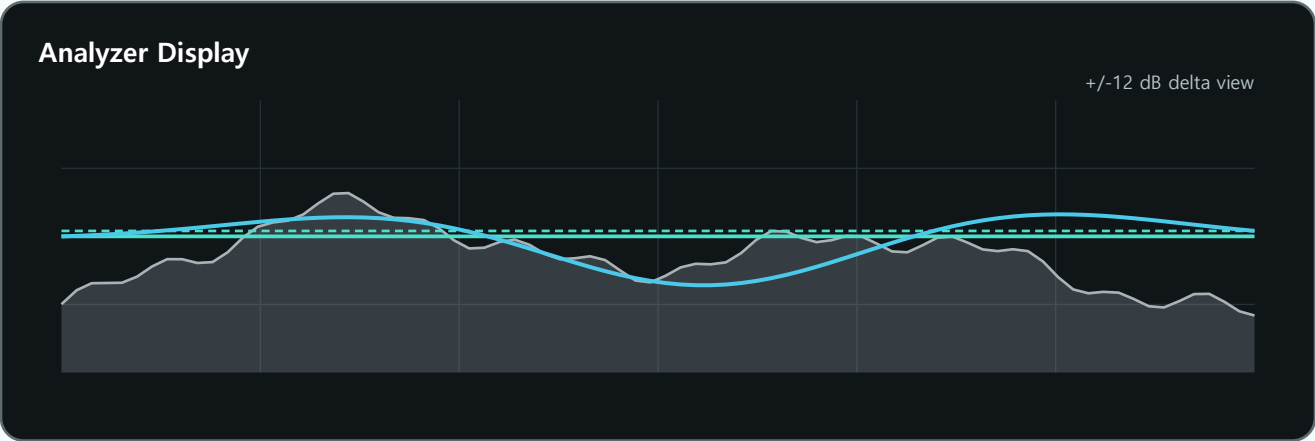
Lookahead is only used as limiter lookahead when the Limiter is enabled.

The plugin has two operating modes. Simple Mode applies one sensitivity line across the whole spectrum. EQ Mode builds separate frequency-dependent sensitivity curves for Attack and Sustain using Low Shelf, Bell, and High Shelf bands.

Quick Start

- Insert the plugin and leave Input Gain and Output Gain at 0 dB.
- Start in Simple Mode. In each analyzer, the horizontal sensitivity line is the detection threshold. Near the top is close to pass-through; lower positions detect more signal.
- For kick or snare impact, lower the Attack sensitivity line until the attack delta reacts, then adjust Attack Gain.
- For body or tail shaping, lower the Sustain sensitivity line until the sustain delta reacts, then adjust Sustain Gain.
- Turn on Saturation, Clipper, or Limiter only when needed. Enabled buttons glow in a yellow/orange lamp color.
- For peak control, enable the Limiter and lower Threshold. Lookahead only matters while the Limiter is on.

Interface



Area	Purpose	Operation
Simple Mode	Detects the whole spectrum with one sensitivity line	Drag the horizontal line inside the analyzer. Higher means less detection, lower means more detection.
EQ Mode	Builds per-frequency sensitivity with Low Shelf, Bell and High Shelf filters.	Drag and hold shift/ctrl for frequency and up/down for sensitivity. Mouse wheel for frequency and up/down for sensitivity.
Attack Spectrum	Shows fast rising transient activity and the applied delta	Only delta means the attack stage is currently changing the signal.
Sustain Spectrum	Shows stable energy after the attack and the applied delta	Only delta means the sustain stage is currently changing the signal.
Global Grid	Controls Oversampling, Lookahead, Saturation, Clipping, and Input/Output Gain.	Click on single, when enabled.

Reading The Analyzer

- The main spectrum is the input source, not the delta curve.
- The delta curve shows the actual applied change. The visual range is +/-12 dB.
- When there is signal but no applied change, a colored 0 dB line is shown. The silent 0 Hz region is not drawn.
- When Attack or Sustain activity stops, the analyzer display releases smoothly over roughly 50 ms.

Modes And Detection

Simple Mode

In Simple Mode, the horizontal sensitivity line in each analyzer is the detection threshold for Attack or Sustain. At the very top, normalized sensitivity is near zero and transient shaping is effectively bypassed. Moving the line downward lowers the threshold and makes smaller signals eligible for processing.

EQ Mode

In EQ Mode, all bands start at the centered neutral value of 0.5. Low is a Low Shelf, Mid is a Bell, and High is a High Shelf. At the center value the curve is close to flat; moving a node vertically changes the detection threshold in that frequency region.

Attack / Sustain

- Attack is fast-rising energy compared with the previous frame. It uses dB-domain spectral-flux-style delta logic.
- Sustain is stable energy after an attack and before the next attack. It uses the same sensitivity concept, but favors continuity and stability instead of sudden rise.
- Attack Gain and Sustain Gain are dB gain sliders applied only to detected regions. Their range is -24 dB to +24 dB.
- Internal smoothing prevents abrupt jumps to the target gain, reducing clicks and unstable motion.

Control Reference

Control	Range / Choices	Default	Description
Mode	Simple / EQ	Simple	Simple uses a full-band line; EQ uses three frequency bands.
Attack Gain	-24 dB to +24 dB	0 dB	Gain applied to detected attack content.
Sustain Gain	-24 dB to +24 dB	0 dB	Gain applied to detected sustain content.
Attack Sensitivity	0.0 to 1.0	0.125	Simple Mode default corresponds to roughly a -10 dB detection point.
Sustain Sensitivity	0.0 to 1.0	0.125	Simple Mode default corresponds to roughly a -10 dB detection point.
EQ Band Frequency	20 Hz to 20 kHz	100 / 1k / 8k	Low, Mid, and High reference frequencies.
EQ Band Sensitivity	0.0 to 1.0	0.5	Centered neutral point for EQ Mode.
EQ Band Q	0.1 to 10.0	1.0	Affects Bell width or Shelf slope.
Oversampling	x1 / x2 / x4 / x8	x1	Applies around the nonlinear Saturation and Clipper stage.
Lookahead	0 ms to 20 ms	0 ms	Used only by the Limiter while it is enabled.
Saturation	On / Off	Off	Enables saturation Drive and Mode.
Saturation Drive	0 dB to 36 dB	0 dB	Input drive into the saturation stage.
Saturation Mode	Soft / Punch / Warm / Digital / Soft / Hard	Soft	Tone model for saturation.
Clipper	On / Off	Off	Enables threshold-based clipping.
Clipper Threshold	-20 dB to 0 dB	-0.7 dB	Clipper operating level.
Limiter	On / Off	Off	Final peak control.
Limiter Threshold	-24 dB to 0 dB	-0.1 dB	Limiter ceiling target.
Input Gain	-24 dB to +24 dB	0 dB	Applied before transient analysis.
Output Gain	-24 dB to +24 dB	0 dB	Applied after nonlinear processing and before the limiter.

Reset And Workflow Tips

Action	Result
Double-click a slider	Resets that slider to its default value.
Alt + click a slider	Resets that slider to its default value.
Double-click an EQ node	Resets that node frequency, sensitivity, and Q.
Alt + click an EQ node	Same reset behavior as double-click.

Practical Tips

- If transient shaping does nothing, first check whether the sensitivity line is too high. The top position is close to pass-through.
- If Attack and Sustain delta move together too much, raise the Attack threshold slightly and tune Sustain separately.
- When using Saturation or Clipper, x2 or higher Oversampling can reduce high-frequency aliasing. Return to x1 if CPU load matters more.
- After rebuilding the plugin, many DAWs need the plugin instance reloaded. Some hosts lock the VST3 file until the DAW is closed.
- If an older session shows an odd Clipper Threshold value, reset that control once.

Default State

On first load, Saturation, Clipper, and Limiter are disabled. Input/Output Gain, Attack/Sustain Gain, Saturation Drive, and Lookahead start at 0. Simple Mode Attack/Sustain Sensitivity starts around a -10 dB detection point; EQ Mode bands start at the centered 0.5 sensitivity value.